Field	Value
Document No.	WI-002
Title	CNC Lathe Setup and Operation
Revision	B
Effective Date	2026-08-10
Prepared By	Owner
Approved By	Jacob Honick, Owner
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WI-002 CNC Lathe Setup and Operation

1. Scope and Confirmed Setup

Applies to CNC-LATHE-001, the CTMACH CQ6133 CNC lathe, in the single Torchworx LLC shop. Tool changes are manual. No probing or automatic tool-setting capability is used. The owner confirms the actual tooling, workholding, control functions and inspection methods for each job; this instruction does not imply an automatic turret, powered chuck or other unconfirmed feature.

2. Preparation and Safety

1. Review the current drawing, purchase order, QF-004 traveler and operation/inspection plan. Confirm that the job is within owner-confirmed machining and measurement capability.
2. Check machine condition and applicable maintenance under QP-019. Resolve conditions affecting safety or conformity before use.
3. Clean the workholding and tool-mounting surfaces. Select suitable available workholding and tools for the job. Verify secure grip, stock support, projection and safe clearance.

4. Wear suitable eye protection, secure hair and loose clothing, and keep hands clear of the rotating workpiece. Remove chuck keys and setup tools before starting. Stop spindle and motion before tool changes, measurement or chip removal; use suitable tools to handle chips. Use the actual machine safeguards and never bypass them.

3. Setup and Verification

1. Mount the stock and verify alignment/runout by a suitable method against the actual job requirements. Confirm grip without distorting the part.
2. Install each cutting tool manually with the spindle stopped. Establish tool positions and work datums manually using an owner-confirmed method suitable for the control and setup. Verify offsets and compensation before cutting.
3. Load the correct program and revision under WI-004. Confirm manual tool-change stops and instructions match the setup.
4. At each tool change, stop spindle and motion, install and secure the correct tool, establish or verify the required offsets, remove setup tools and verify clearance before resuming.
5. Prove out new or changed programs using the actual control functions and conservative motion settings.
6. Produce and fully inspect the first piece under WI-005 before continuing.

4. Production and Closeout

1. Monitor tool condition, workholding, chip evacuation and cutting behavior. Use cutting fluid where suitable for the actual setup.
2. Full inspection is the default under QP-006 and QP-007. Use a different extent only when the customer specifies or approves it in writing and the plan is recorded in the job file. Record results on QF-005 and QF-006 as applicable.
3. Stop for abnormal conditions, workholding movement or nonconformance. Hold affected product under QP-008, correct the cause, and repeat setup/first-piece verification before continuing.
4. Sign off the operation on QF-004, clean the machine safely and record maintenance needs. Use WI-008 status tags.

5. References

- QP-005 Production Planning and Control; QP-006 In-Process Inspection; QP-007 Final Inspection and Product Release
- QP-008 Nonconforming Product Control; QP-019 Preventive Maintenance

- QF-004 Job Traveler; QF-005 In-Process Inspection Record; QF-006 Final Inspection Report; QF-027 Equipment List
- WI-004 CNC Program Verification; WI-005 First Piece Inspection; WI-008 Part Marking

6. Revision History

Rev	Date	Description	Approved By
A	2026-08-06	Initial release	Shop Owner

Revision B Approval

Approved 2026-11-05 by Jacob Honick, Owner. Reflect the CTMACH CQ6133 lathe, manual tool changes and owner-confirmed setup capabilities.